

# PSID Generic Crisis Module

## Master Survey Questionnaire

Optimized via NLP-Driven Utility–Burden Analysis

April 2026

| Field                     | Value                                                                               |
|---------------------------|-------------------------------------------------------------------------------------|
| Architecture              | Generic Core + Crisis-Specific Toggles                                              |
| Generic Core Questions    | 18                                                                                  |
| Pandemic/Disaster Toggle  | 14 (5 pandemic + 5 disaster + 4 trauma)                                             |
| Financial Crisis Toggle   | 3                                                                                   |
| War/Conflict Template     | 4 (placeholder)                                                                     |
| Total Questions           | 39                                                                                  |
| Estimated Duration (Core) | 12–14 minutes                                                                       |
| Time Budget (Max)         | 30 minutes                                                                          |
| Data Sources              | 5 (PSID COVID-19, Katrina 2007, Govt Shutdown, Understanding Society, PSID Generic) |
| Optimization Method       | $R_i = U_i / B_i$ (NLP pipeline with TF-IDF, cosine similarity, co                  |

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## 1. Scripting Notes

This document specifies the PSID Generic Crisis Module questionnaire, an NLP-optimized instrument designed for rapid deployment across diverse crisis contexts. The two-tier architecture separates always-applicable items (Generic Core) from crisis-type-specific toggle modules.

The following modules are deployed in this order within the questionnaire:

- **Generic Core Module** – Always administered (18 questions, approximately 12–14 minutes).
- **Crisis-Specific Toggle Module** – Activated based on crisis type classification. Only one toggle is active per administration.

| Toggle Module               | Activation Condition                                  | Questions |
|-----------------------------|-------------------------------------------------------|-----------|
| Pandemic / Disaster         | Pandemic, epidemic, natural disaster                  | 14 items  |
| Financial Crisis / Shutdown | Government shutdown, recession, banking crisis        | 8 items   |
| War / Armed Conflict        | Armed conflict, civil unrest, displacement (refugees) | 10 items  |

Toggle selection is driven by the crisis classification variable:

**cr\_crisis\_type [Type of crisis event]**

- 1 = Pandemic / Public health emergency
- 2 = Natural disaster (hurricane, flood, earthquake, wildfire)
- 3 = Government shutdown / financial crisis
- 4 = War / armed conflict

## 2. Module Architecture Overview

**Tier 1 (Generic Core):** 18 always-administered questions covering demographics, employment, finances, and wellbeing. Administered to all respondents regardless of crisis type. Estimated time: 12–14 minutes.

**Tier 2 (Crisis-Specific Toggles):** Mutually exclusive modules (14, 3, or 4 items) activated by crisis type. Only one toggle module is administered per respondent.

### Optimization Formula:

$$R_i = U_i / B_i$$

$$B_i = \max(0.10 \times \text{word\_count} + 0.20 \times \text{complexity}, 0.1)$$

$$\text{Enhanced Priority Score: } P_i = \text{Augmented\_Utility} / (B_i \times (1 + 0.65 \times \text{redundancy\_scaled}))$$

$$\text{Augmented\_Utility} = U_i \times (1 + 0.32 \times \text{idf\_scaled} + 0.24 \times \text{construct\_scaled} + 0.12 \times \text{richness\_scaled} + \text{portability\_bonus})$$

**Time Budget:** Maximum 30 minutes. Reading rate: 7 seconds per word. Selected 28 questions total 29.17 minutes.

### 3. Generic Core Module (18 Questions)

#### 3.A Demographics and Household Context

##### **cr\_age [Respondent age]**

*Universe:* Ask all.

*Source:* Understanding Society / PSID baseline

*Scripting notes:* Do not display DK and REF. Range [16-110].

*NLP metrics:*  $R_i = 1.67$  | *Construct:* Demographics | *Always-on core*

What is the respondent's age?

[Numeric textbox: Enter age in years]

##### **cr\_sex [Respondent sex]**

*Universe:* Ask all.

*Source:* Understanding Society covid-19 survey

*Scripting notes:* Do not display DK and REF, answer required.

*NLP metrics:*  $R_i = 1.23$  | *Construct:* Demographics | *Always-on core*

And are you...

1. Male
2. Female
3. Another identity or prefer not to say

##### **cr\_addrchk [Address check]**

*Universe:* Ask all.

*Source:* Understanding Society covid-19 survey

*NLP metrics:*  $R_i = 1.17$  | *Construct:* Demographics | *Always-on core*

Can I just check, are you normally resident at this address?

1. Yes
2. No

##### **cr\_hhsize [Household size]**

*Universe:* Ask all.

*Source:* PSID Generic Crisis Module (expanded)

*Scripting notes:* Range [1-20]. Answer required.

*NLP metrics:* Expanded generic item | *Construct:* Demographics

How many people currently live in your household, including yourself?

[Numeric textbox]

##### **cr\_depend [Dependents]**

*Universe:* Ask all.

*Source:* PSID Generic Crisis Module (expanded)

*NLP metrics:* Expanded generic item | Household vulnerability context

Do you have any children or dependents currently living with you?

1. Yes
2. No

### **3.B Employment and Income Disruption**

#### **cr\_empstat [Employment status]**

*Universe:* Ask all.

*Source:* PSID Generic Crisis Module (expanded)

*Scripting notes:* Single-code. Answer required.

*NLP metrics:* Expanded generic item | Construct: Employment

What is your current employment status?

1. Employed full-time
2. Employed part-time
3. Self-employed
4. Temporarily laid off
5. Unemployed, looking for work
6. Unemployed, not looking for work
7. Retired
8. Unable to work (disability)
9. Student
10. Other (please specify)

#### **cr\_stopwork [Stopped work]**

*Universe:* IF cr\_empstat = employed or self-employed in previous wave.

*Source:* PSID Government Shutdown Income module

*NLP metrics:*  $R_i = 4.67$  | Construct: Employment | Always-on core

Have you stopped this work because of the crisis?

1. Yes
2. No
3. Not applicable

#### **cr\_stopbus [Stopped business]**

*Universe:* IF self-employed or business owner in previous wave.

*Source:* PSID Government Shutdown Income module

*NLP metrics:*  $R_i = 4.50$  | Construct: Economic / Income | Always-on core

Have you stopped working at this business because of the crisis?

1. Yes
2. No
3. Not applicable

### **cr\_wages [Wages during crisis]**

*Universe:* IF cr\_empstat = employed.

*Source:* PSID Government Shutdown Income module

*NLP metrics:*  $R_i = 2.73$  | *Construct:* Economic / Income | Always-on core

**Were there any wages or salary payments from this job during the crisis period?**

1. Yes
2. No
3. Don't know

### **cr\_hourschange [Change in work hours]**

*Universe:* IF cr\_empstat = employed or self-employed.

*Source:* PSID Generic Crisis Module (expanded)

*NLP metrics:* Expanded generic item | *Construct:* Employment

**Did your working hours change as a result of the crisis?**

1. Increased significantly
2. Increased somewhat
3. Stayed about the same
4. Decreased somewhat
5. Decreased significantly

### 3.C Financial Coping and Public Support

#### cr\_findiff [Financial difficulties]

Universe: Ask all.

Source: PSID COVID-19 module

NLP metrics:  $R_i = 5.67$  | Construct: Financial Coping | Always-on core (HIGHEST RANKED)

Have you experienced any financial difficulties because of the crisis?

1. Yes
2. No

#### cr\_govtaid [Government assistance]

Universe: Ask all.

Source: PSID Generic Crisis Module (documentation-backed)

NLP metrics: Documentation-backed generic item | Construct: Government Aid

Did you receive any government financial assistance because of the crisis?

1. Yes
2. No
3. Don't know

#### cr\_copestrategy [Coping strategy]

Universe: IF cr\_findiff = Yes.

Source: PSID Generic Crisis Module (expanded)

NLP metrics: Expanded generic item | Construct: Financial Coping

How did your household manage these financial difficulties? (Select all that apply)

1. Used savings
2. Borrowed from family or friends
3. Sold belongings or assets
4. Took on additional debt (credit card, loan)
5. Reduced spending on food or essentials
6. Took on a second job or extra work
7. Used a food bank or community support
8. Delayed paying bills (rent, mortgage, utilities)
9. Other (please specify)

#### cr\_incomeloss [Income loss estimate]

Universe: IF cr\_findiff = Yes.

Source: PSID Generic Crisis Module (expanded)

NLP metrics: Expanded generic item | Construct: Economic / Income

Approximately how much did your household's total income decrease as a result of the crisis?

1. Less than 10%



2. 10–25%
3. 26–50%
4. More than 50%
5. Don't know

### **cr\_recovery [Financial recovery outlook]**

*Universe:* IF cr\_findiff = Yes.

*Source:* PSID Generic Crisis Module (expanded)

*NLP metrics:* Expanded generic item | *Construct:* Financial Coping

How long do you expect it will take for your household's finances to return to pre-crisis levels?

1. Already recovered
2. Less than 6 months
3. 6 months to 1 year
4. 1–2 years
5. More than 2 years
6. Don't expect to recover

## **3.D Housing Stability and Wellbeing**

### **cr\_housing [Housing instability]**

*Universe:* Ask all.

*Source:* PSID Generic Crisis Module (documentation-backed)

*NLP metrics:* Documentation-backed | *Construct:* Housing / Shelter, Economic / Income

Did your household fall behind on rent, mortgage, utilities, or other major bills because of the crisis?

1. Yes
2. No
3. Not applicable (no such obligations)

### **cr\_wellbeing [Wellbeing check]**

*Universe:* Ask all.

*Source:* PSID Generic Crisis Module (documentation-backed)

*NLP metrics:* Documentation-backed | *Construct:* Trauma / Health

Since the crisis, how often have you felt worried, unable to relax, or unable to sleep well?

1. Not at all
2. Several days
3. More than half the days
4. Nearly every day

### **cr\_childimpact [Impact on children]**

*Universe:* IF cr\_depend = Yes.

*Source:* PSID Generic Crisis Module (expanded, CDS/SDQ items)

*NLP metrics:* Expanded generic item | *Construct:* Trauma / Health

**Has the crisis noticeably affected the emotional wellbeing or behaviour of any children in your household?**

1. Yes, significantly
2. Yes, somewhat
3. No
4. Don't know

## 4. Crisis-Specific Toggle: Pandemic / Disaster

*Activated when cr\_crisis\_type = 1 or 2.*

### 4.A Pandemic Economic Impact (5 questions)

#### **pd\_earningloss [Pandemic earnings loss]**

*Universe:* Ask if crisis type = pandemic.

*Source:* PSID COVID-19 module

*NLP metrics:*  $R_i = 5.67$  | *Construct:* Economic / Income | *Documentation-backed*

Did you lose earnings because of the pandemic?

1. Yes
2. No

#### **pd\_stimulus [Stimulus payment]**

*Universe:* Ask if crisis type = pandemic.

*Source:* PSID COVID-19 module

*NLP metrics:*  $R_i = 4.67$  | *Construct:* Government Aid | *Documentation-backed*

Did you receive a stimulus payment or other emergency government support?

1. Yes
2. No

#### **pd\_essential [Essential work]**

*Universe:* Ask if crisis type = pandemic AND cr\_empstat = employed.

*Source:* PSID COVID-19 module

*NLP metrics:*  $R_i = 4.00$  | *Construct:* Employment | *Documentation-backed*

Were you working in a job that was considered essential during the crisis?

1. Yes
2. No

#### **pd\_wfh [Work from home]**

*Universe:* Ask if crisis type = pandemic AND cr\_empstat = employed.

*Source:* PSID COVID-19 module

*NLP metrics:*  $R_i = 3.50$  | *Construct:* Employment

Did you work entirely from home during the crisis?

1. Yes
2. No
3. Partially (mix of home and on-site)

#### **pd\_layoff [Layoff/furlough]**

*Universe:* Ask if crisis type = pandemic.

*Source:* PSID COVID-19 module

*NLP metrics:*  $R_i = 2.32$  | *Construct:* Employment

Were you laid off or furloughed because of the pandemic?

1. Yes
2. No

## 4.B Disaster Housing and Displacement (5 questions)

### tr\_evacuate [Evacuation]

*Universe:* Ask if crisis type = natural disaster.

*Source:* PSID Hurricane Katrina 2007 module

*NLP metrics:*  $R_i = 3.18$  | *Construct:* Housing / Shelter | *Documentation-backed*

Did you evacuate from your home before the disaster hit?

1. Yes
2. No

### tr\_flooding [Home flooding]

*Universe:* Ask if crisis type = natural disaster.

*Source:* PSID Hurricane Katrina 2007 module

*NLP metrics:*  $R_i = 2.68$  | *Construct:* Housing / Shelter

Did you experience major flooding in your home during the disaster?

1. Yes
2. No

### tr\_damage [Property damage]

*Universe:* Ask if crisis type = natural disaster.

*Source:* PSID Hurricane Katrina 2007 module

*NLP metrics:*  $R_i = 2.53$  | *Construct:* Housing / Shelter | *Documentation-backed*

How severe was the damage to your home during the disaster?

1. None / minimal
2. Minor (cosmetic or easily repaired)
3. Moderate (structural repair needed)
4. Severe (uninhabitable)
5. Destroyed

### tr\_temphousing [Temporary housing]

*Universe:* Ask if tr\_evacuate = Yes OR tr\_damage >= Moderate.

*Source:* PSID Hurricane Katrina 2007 module

*NLP metrics:*  $R_i = 1.59$  | *Construct:* Housing / Shelter | *Documentation-backed*

How long did you stay in temporary housing after the disaster?

1. Did not need temporary housing
2. Less than 1 week
3. 1–4 weeks
4. 1–6 months
5. More than 6 months

**tr\_returnhome [Return to home]**

*Universe:* Ask if tr\_evacuate = Yes OR tr\_damage >= Moderate.

*Source:* PSID Hurricane Katrina 2007 module

*NLP metrics:*  $R_i = 1.70$  | *Construct:* Housing / Shelter

Have you been able to return to your original home?

1. Yes, fully returned
2. Yes, but home still under repair
3. No, living elsewhere permanently
4. No, still in temporary housing

## 4.C Trauma, Injury, and Exposure (4 questions)

### tr\_afraid [Fear during disaster]

*Universe:* Ask if crisis type = natural disaster.

*Source:* PSID Hurricane Katrina 2007 module

*NLP metrics:*  $R_i = 1.68$  | *Construct:* Trauma / Health

How afraid were you during the disaster that you might be seriously injured or killed?

1. Not at all afraid
2. Slightly afraid
3. Moderately afraid
4. Very afraid
5. Extremely afraid

### tr\_injured [Physical injury]

*Universe:* Ask if crisis type = natural disaster.

*Source:* PSID Hurricane Katrina 2007 module

*NLP metrics:*  $R_i = 1.45$  | *Construct:* Trauma / Health

Were you physically injured in any way as a result of the disaster?

1. Yes
2. No

### tr\_fema [FEMA assistance]

*Universe:* Ask if crisis type = natural disaster AND country = US.

*Source:* PSID Hurricane Katrina 2007 module

*NLP metrics:*  $R_i = 2.09$  | *Construct:* Government Aid

Did you receive any help from FEMA?

1. Yes
2. No
3. Did not apply

### tr\_distress [Post-disaster distress]

*Universe:* Ask if crisis type = natural disaster.

*Source:* PSID Hurricane Katrina 2007 module (condensed)

*Scripting notes:* Grid format with frequency scale.

*NLP metrics:* Condensed from multiple Katrina items | *Construct:* Trauma / Health

Since the disaster, how often have you experienced the following? [Grid format with frequency scale: Not at all / Several days / More than half the days / Nearly every day]

1. Repeated disturbing memories or images
2. Disturbing dreams about the event
3. Feeling nervous, anxious, or on edge

#### 4. Trouble relaxing

## 5. Alternate Toggle: Financial Crisis / Shutdown

*Activated when cr\_crisis\_type = 3.*

### **fc\_manage [Financial difficulty management]**

*Universe:* Ask if crisis type = financial crisis / government shutdown.

*Source:* PSID Government Shutdown Crisis module

*Scripting notes:* Multi-code. Select all that apply.

*NLP metrics:*  $R_i = 3.72$  | *Construct:* Financial Coping | *Documentation-backed*

How did your household manage financial difficulties caused by the shutdown?

1. Used savings
2. Borrowed from family or friends
3. Sold belongings
4. Took on debt
5. Reduced spending
6. Other

### **fc\_paycheck [Missed paycheck]**

*Universe:* IF cr\_empstat = employed.

*Source:* PSID Government Shutdown prototype extension

*NLP metrics:* *Construct:* Economic / Income

Did your household miss a paycheck or salary payment because of the shutdown?

1. Yes
2. No
3. Not applicable

### **fc\_backpay [Back pay expectation]**

*Universe:* IF fc\_paycheck = Yes.

*Source:* PSID Government Shutdown prototype extension

*NLP metrics:* *Construct:* Government Aid

Did you expect to receive back pay or repayment later?

1. Yes
2. No
3. Unsure



## 6. Template Toggle: War / Armed Conflict

*Activated when cr\_crisis\_type = 4. These questions are placeholders for future development – not yet NLP-scored.*

### **wc\_displace [Conflict displacement]**

*Universe:* Ask if crisis type = armed conflict.

*Source:* PSID War/Conflict template (placeholder)

*NLP metrics:* Template | Construct: Housing / Shelter / Trauma

Were you forced to leave your home because of armed conflict or security concerns?

1. Yes
2. No

### **wc\_workint [Conflict work interruption]**

*Universe:* Ask if crisis type = armed conflict.

*Source:* PSID War/Conflict template (placeholder)

*NLP metrics:* Template | Construct: Employment / Economic-Income

Did conflict interrupt your work, business activity, or income source?

1. Yes
2. No
3. Not applicable

### **wc\_violence [Violence exposure]**

*Universe:* Ask if crisis type = armed conflict.

*Source:* PSID War/Conflict template (placeholder)

*NLP metrics:* Template | Construct: Trauma / Health

Did you or anyone in your immediate family experience injury, detention, or threats of violence?

1. Yes
2. No
3. Prefer not to say

### **wc\_agency [Agency assistance]**

*Universe:* Ask if crisis type = armed conflict.

*Source:* PSID War/Conflict template (placeholder)

*NLP metrics:* Template | Construct: Government Aid

Did you receive assistance from government, humanitarian, or refugee-support agencies?

1. Yes
2. No
3. Not applicable

## 7. NLP Optimization Methodology Summary

This module was developed using a dual-pipeline NLP approach to maximize utility while minimizing respondent burden. The process began with 52 candidate questions extracted from five PSID historical crisis surveys: COVID-19 (19 items), Hurricane Katrina 2007 (18 items), Government Shutdown Income (8 items), Understanding Society (4 items), and generic PSID modules (3 items). The NLP pipeline employed RAKE (Rapid Automatic Keyword Extraction) keyphrase extraction and spaCy noun-phrase chunking to identify and map candidate question concepts to a 63-entry crisis taxonomy spanning seven constructs: Demographics, Employment, Economic/Income, Financial Coping, Housing/Shelter, Government Aid, and Trauma/Health. Each question was scored using the utility-burden formula  $R_i = U_i / B_i$ , where utility was derived from TF-IDF scores, cosine similarity to construct vocabularies, and construct richness metrics. Burden incorporated word count, linguistic complexity, and redundancy penalties. An enhanced priority score  $P_i$  further weighted portable and well-attested items. The selection algorithm proceeded in two stages: first, all 18 generic core questions were selected to ensure comprehensive coverage of always-applicable domains. Second, crisis-specific items were ranked and selected within a maximum 30-minute respondent time budget, capped at 7 seconds per word of reading time. Final result: 28 selected questions (18 generic core + 10 toggle items) requiring approximately 29.17 minutes of respondent time, achieving optimal utility-burden balance for deployment in diverse crisis contexts.

## 8. Appendix — Ranked Question Summary Table

| #  | Question                                                   | Source                 | Toggle            | Rank | Selected |
|----|------------------------------------------------------------|------------------------|-------------------|------|----------|
| 1  | Any financial difficulties                                 | COVID-19               | Generic Core      | 5.67 | Yes      |
| 2  | Lost earnings because of the pandemic                      | COVID-19               | Pandemic/Disaster | 5.67 | Yes      |
| 3  | Stopped this work?                                         | Govt Shutdown Income   | Generic Core      | 4.67 | Yes      |
| 4  | Received stimulus payment                                  | COVID-19               | Pandemic/Disaster | 4.67 | Yes      |
| 5  | Stopped working at this business?                          | Govt Shutdown Income   | Generic Core      | 4.50 | Yes      |
| 6  | Working in a job that was considered essential work?       | COVID-19               | Pandemic/Disaster | 4.00 | Yes      |
| 7  | How did you manage financial difficulties due to shutdown? | Govt Shutdown Crisis   | Financial Crisis  | 3.72 | Yes      |
| 8  | Only work from home                                        | COVID-19               | Pandemic/Disaster | 3.50 | Yes      |
| 9  | Did you evacuate from your home before the disaster?       | Hurricane Katrina 2007 | Pandemic/Disaster | 3.18 | Yes      |
| 10 | Wages or salary from this job                              | Govt Shutdown Income   | Generic Core      | 2.73 | Yes      |
| 11 | Major flooding in your home                                | Hurricane Katrina 2007 | Pandemic/Disaster | 2.68 | Yes      |
| 12 | Property damage to your home                               | Hurricane Katrina 2007 | Pandemic/Disaster | 2.53 | Yes      |
| 13 | Laid off or furloughed because of the pandemic             | COVID-19               | Pandemic/Disaster | 2.32 | Yes      |
| 14 | Financial help from FEMA                                   | Hurricane Katrina 2007 | Pandemic/Disaster | 2.09 | Yes      |
| 15 | Return to original home                                    | Hurricane Katrina 2007 | Pandemic/Disaster | 1.70 | Yes      |
| 16 | How afraid during the disaster                             | Hurricane Katrina 2007 | Pandemic/Disaster | 1.68 | Yes      |
| 17 | Respondent age                                             | Understanding Society  | Generic Core      | 1.67 | Yes      |
| 18 | Temporary housing after the disaster                       | Hurricane Katrina 2007 | Pandemic/Disaster | 1.59 | Yes      |
| 19 | Are you male/female                                        | Understanding Society  | Generic Core      | 1.23 | Yes      |
| 20 | Normally resident at this address                          | Understanding Society  | Generic Core      | 1.17 | Yes      |

## 9. References

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